Vector Calculus and Complex Analysis Classwork (2025/4/24)

Name:

Student ID:

Solve $\cos z = -2i$

Solution

$$\frac{e^{iz}+e^{-iz}}{2}=-2i$$

$$e^{iz}+4i+e^{-iz}=0$$

$$w^2+4iw+1=0,\quad w=e^{iz}$$

$$e^{iz}=w=\frac{-4i\pm\sqrt{(4i)^2-4}}{2}=\left(\sqrt{5}-2\right)i \quad \text{or} \quad -\left(2+\sqrt{5}\right)i$$

$$iz=\operatorname{Log}\left(\sqrt{5}-2\right)+\left(2k+\frac{1}{2}\right)\pi i \quad \text{or} \quad \operatorname{Log}\left(2+\sqrt{5}\right)+\left(2k-\frac{1}{2}\right)\pi i$$

$$z=\left(2k+\frac{1}{2}\right)\pi-i\operatorname{Log}\left(\sqrt{5}-2\right)=\left(2k+\frac{1}{2}\right)\pi+i\operatorname{Log}\left(\sqrt{5}+2\right) \quad \text{or} \quad \left(2k-\frac{1}{2}\right)\pi-i\operatorname{Log}\left(2+\sqrt{5}\right)$$

$$=\left(2k\pm\frac{1}{2}\right)\pi\pm i\operatorname{Log}\left(\sqrt{5}+2\right),\quad k=0,\pm1,\pm2,\cdots$$